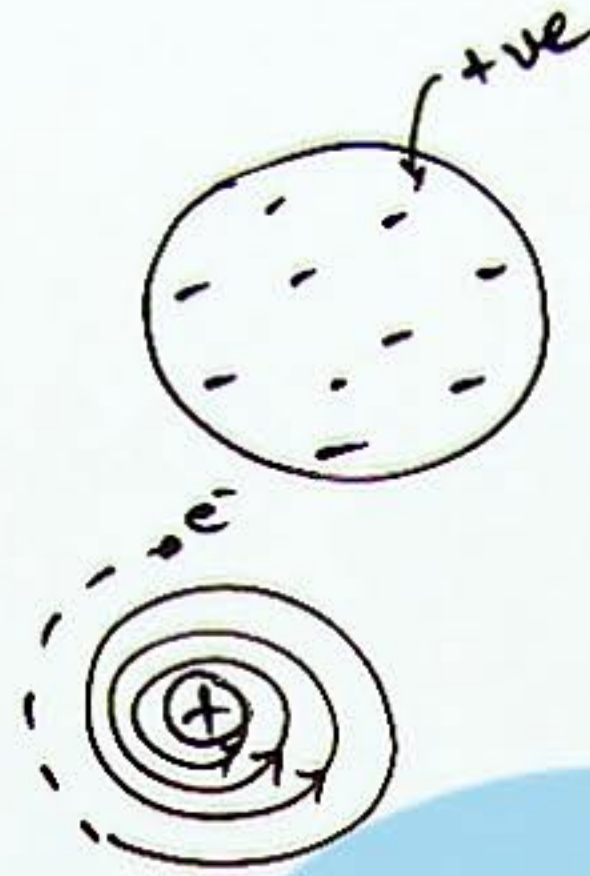


① Structure of Atom :



(i) Thomson Model

(ii) Rutherford Model

(iii) Bohr Model

② Bohr's Atomic Model :

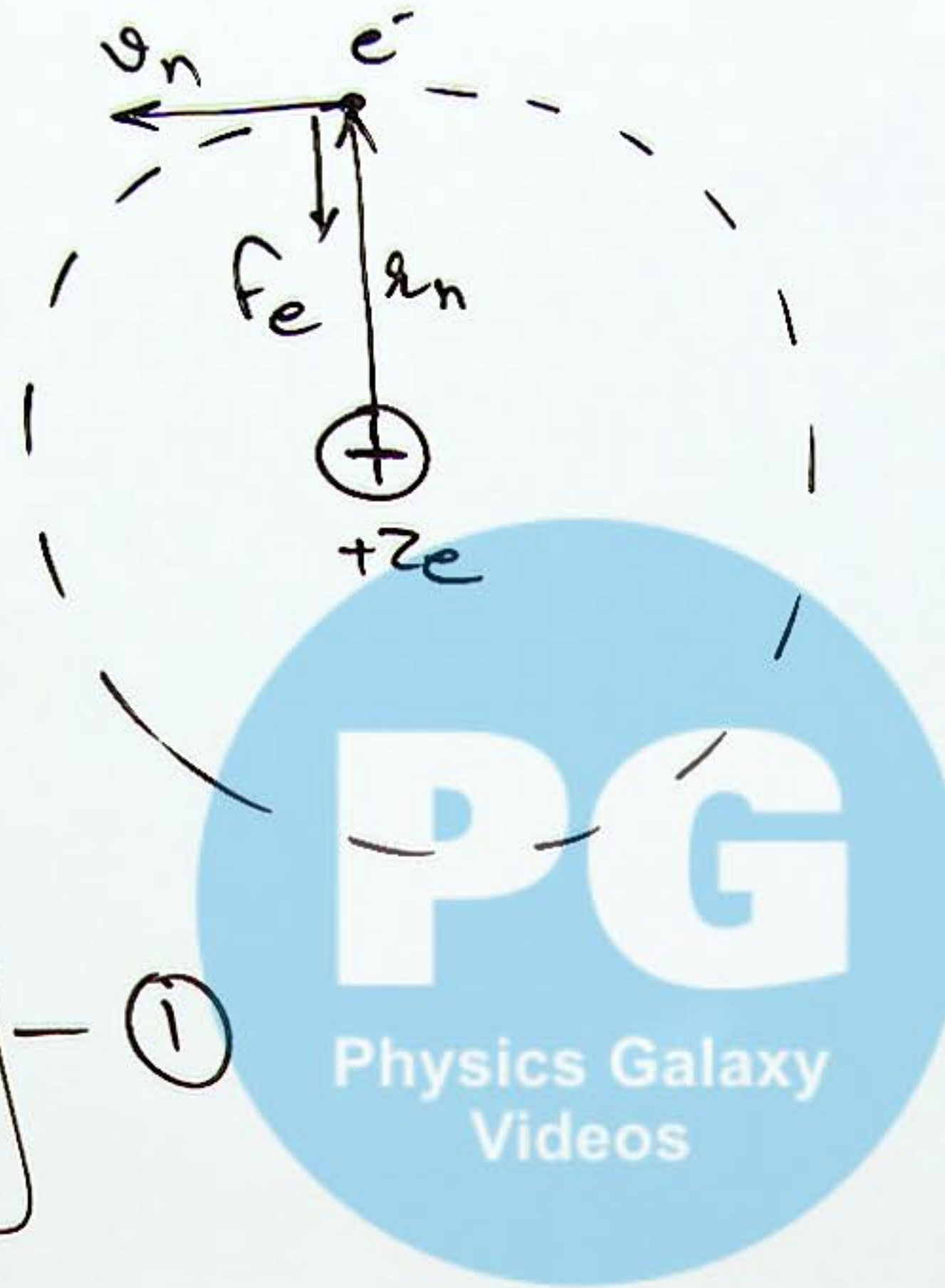
Postulates of Bohr Model

(i) I Postulate

$$f_e = \frac{kq_1q_2}{r^2} = \frac{kZe^2}{r_n^2}$$

$$\frac{kZe^2}{r_n^2} = \frac{mv_n^2}{r_n}$$

$$mv_n^2 = \frac{kZe^2}{r_n}$$



② Bohr's Atomic Model :

Postulates of Bohr Model

(i) I Postulate

(ii) II Postulate

Condition

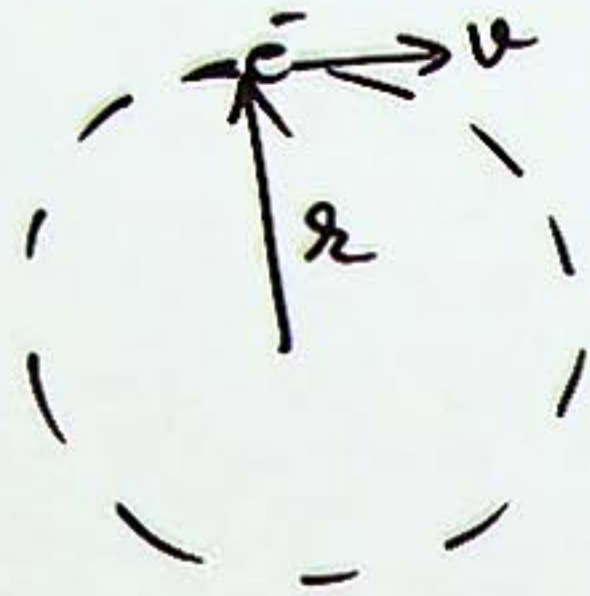
$$\boxed{m v_n r_n = n \left(\frac{h}{2\pi} \right)}$$

②

$n \in \mathbb{I}$

e^- can revolve around Nu only

in some stable orbits



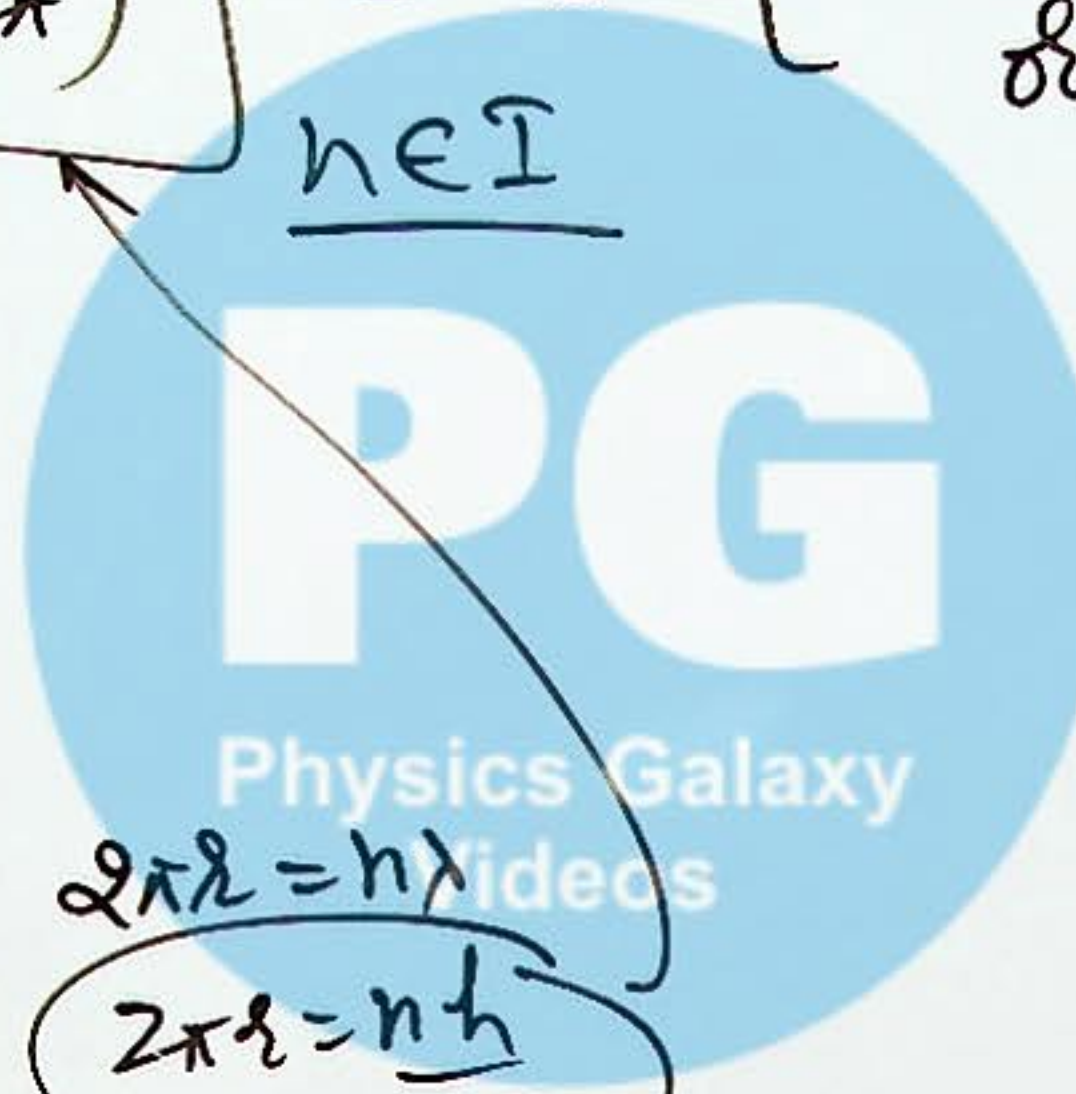
W/L of e^- wave

$$\lambda = \frac{h}{mv}$$

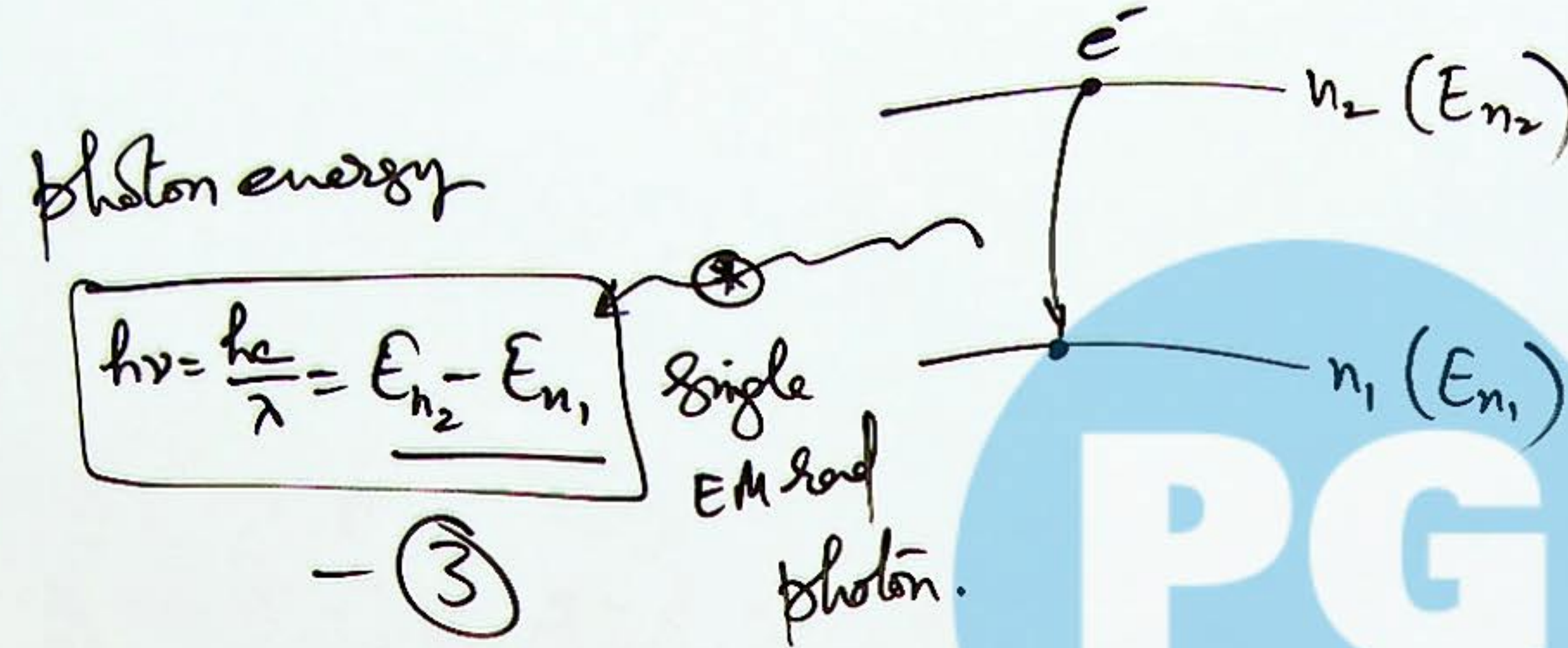
for S/W formation

$$2\pi r = n\lambda$$

$$2\pi r = n \frac{h}{mv}$$



② Bohr's Atomic Model :



Postulates of Bohr Model

- (i) I Postulate
- (ii) II Postulate
- (iii) III Postulate

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③ Properties of e^- in Bohr's Atomic Model:

① $\rightarrow \frac{kZe^2}{r_n} = mv_n^2$; ② $\rightarrow mvr_n r_n = n\left(\frac{h}{2\pi}\right)$; ③ $\rightarrow E_{n_2} - E_{n_1} = h\nu = \frac{hc}{\lambda}$

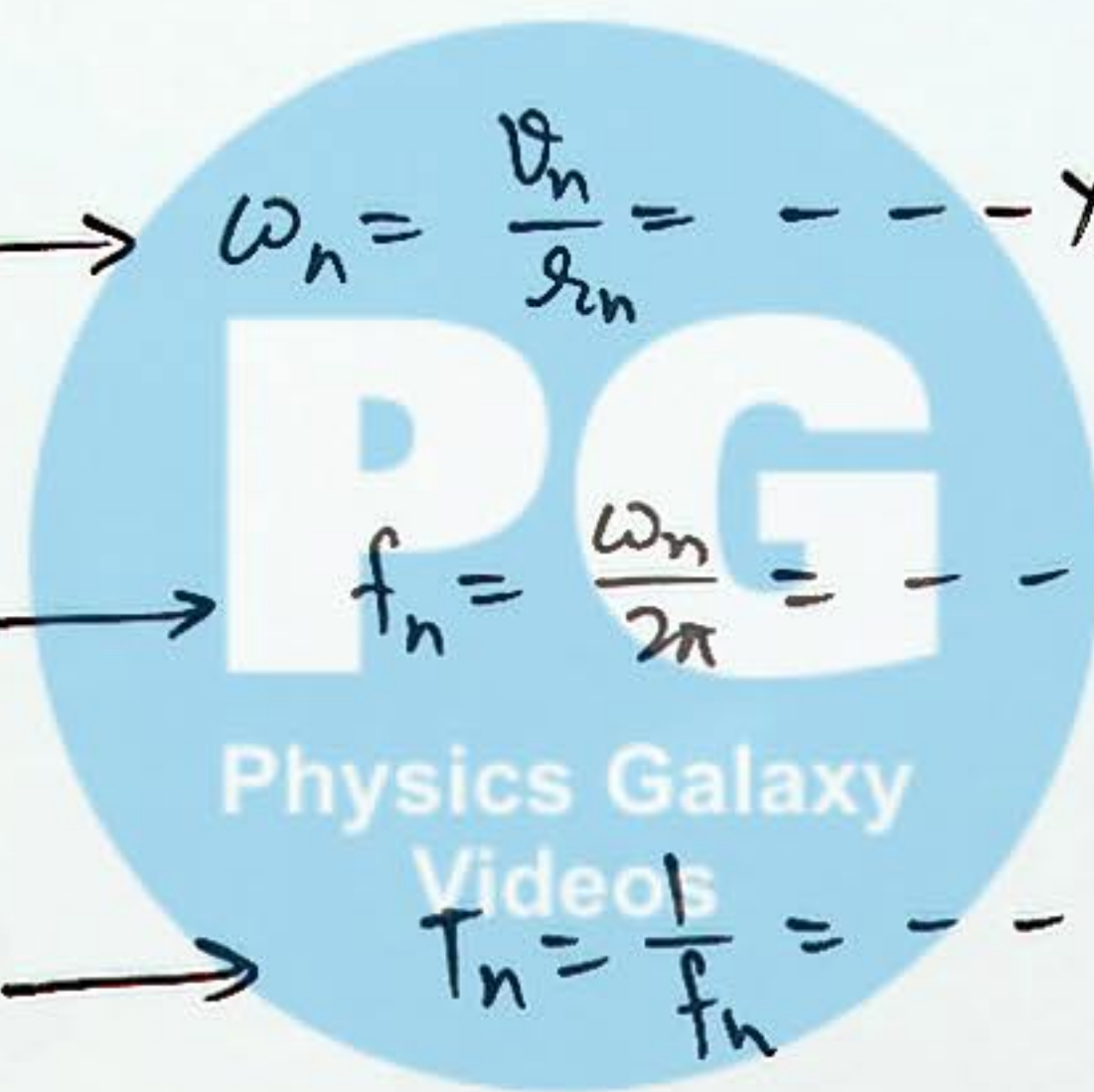
(a) Radius of n^{th} orbit $\rightarrow r_n = \frac{n^2 h^2}{4\pi^2 kZe^2 m_e} = 0.529 \times \frac{n^2}{Z} \text{ \AA}$

(b) Velocity in n^{th} orbit $\rightarrow v_n = \frac{2\pi kZe^2}{nh} = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$

(c) Angular vel in n^{th} orbit $\rightarrow \omega_n = \frac{v_n}{r_n} = \dots \times \frac{Z^2}{n^3} \text{ rad/s}$

(d) Frequency in n^{th} orbit $\rightarrow f_n = \frac{\omega_n}{2\pi} = \dots \times \frac{Z^2}{n^3} \text{ sec}^{-1}$

(e) Time period in n^{th} orbit $\rightarrow T_n = \frac{1}{f_n} = \dots \times \frac{n^3}{Z^2} \text{ sec}$



② Properties of e^- in Bohr's Atomic Model:

$$(P1) \rightarrow \frac{kZe^2}{r_n} = mv_n^2 ; (P2) \rightarrow mvr_n r_n = n\left(\frac{h}{2\pi}\right) ; (P3) \rightarrow E_{n_2} - E_{n_1} = h\nu = \frac{hc}{\lambda}$$

(f) Current in n^{th} Orbit $\rightarrow I_n = \text{charge} \times \text{rev freq.} = e f_n = \dots \times \frac{Z^2}{n^3}$

(g) Magnetic Induction at $N_4 \rightarrow B_n = \frac{\mu_0 I_n}{2r_n} = \dots \times \frac{Z^3}{n^5}$

(h) Magnetic Moment of n^{th} Orbit $\rightarrow M_n = I_n \times \pi r_n^2 = \frac{enh}{4\pi m}$ for $n=1 \rightarrow M_1 = \frac{eh}{4\pi m}$
Bohr Magneton

(i) Energy of e^- in n^{th} orbit $\rightarrow E_n = K_n + P_n$

$$= \frac{1}{2} \frac{kZe^2}{r_n} - \frac{kZe^2}{r_n} = -\frac{1}{2} \frac{kZe^2}{r_n} = -\frac{2\pi^2 k^2 Z^2 e^4 m}{h^2 h^2} = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

④ Energies of different energy levels in a hydrogenic Atom:

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

$$\checkmark E_1 = -13.6 Z^2 \text{ eV}$$

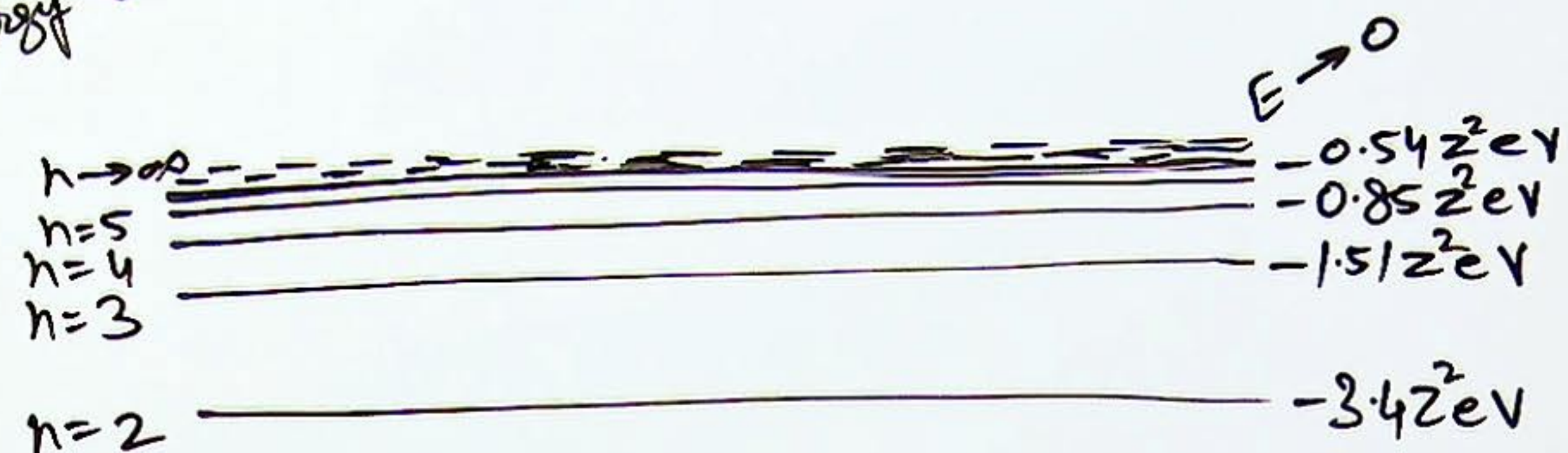
$$E_2 = -\frac{13.6}{2^2} Z^2 = -3.4 Z^2 \text{ eV}$$

$$E_3 = -\frac{13.6}{3^2} Z^2 = -1.51 Z^2 \text{ eV}$$

$$\checkmark E_4 = -\frac{13.6}{4^2} Z^2 = -0.85 Z^2 \text{ eV}$$

$$E_5 = -\frac{13.6}{5^2} Z^2 = -0.54 Z^2 \text{ eV}$$

Energy level diagram



0.66 eV

0.31 eV

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4
1

$$\Delta E_{4 \rightarrow 1} = 12.75 \text{ eV}$$

2
1

$$\Delta E_{2 \rightarrow 1} = 10.2 \text{ eV}$$

3
1
4

$$\Delta E_{3 \rightarrow 1} = 12.09 \text{ eV}$$

2

$$\Delta E_{4 \rightarrow 2} = 2.55 \text{ eV}$$

⑤ Excitation & Ionization of an Atom:

Transition from lower to higher orbit

from any orbit to ejection

transition to $n \rightarrow \infty$

NOTE: If supp energy $>$ IP then excess energy will become KE of ejected e^-

$$\Delta E_{13} = 12.09 \text{ eV}$$

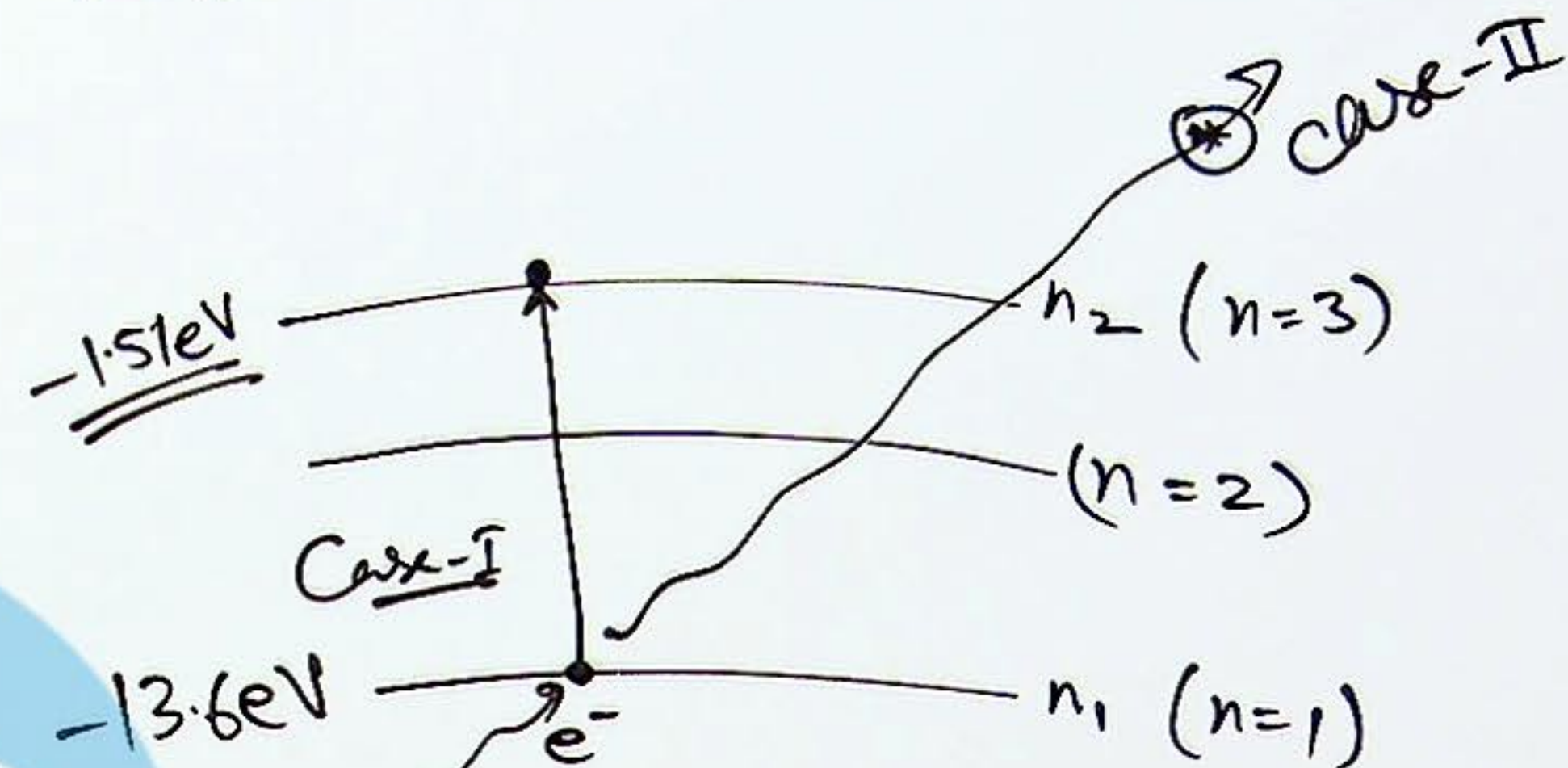
$$\Delta E_{12} = 10.2 \text{ eV}$$

There are two ways in which energy can be supplied for excitation/ionization

① by photon

② by collision

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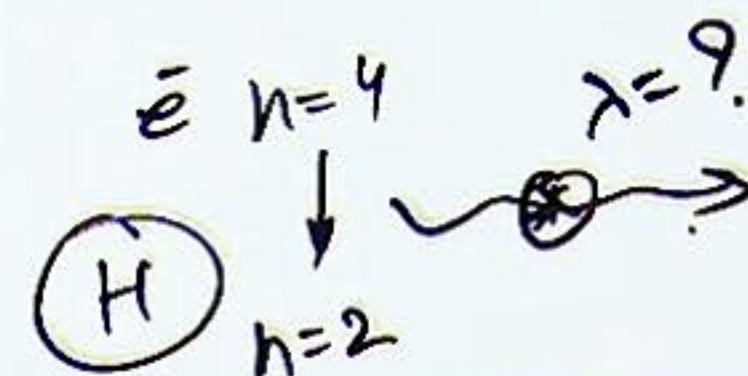
Case-I: $E_{ph} = 12.09 \text{ eV}$

Case-II: $E'_{ph} = 11 \text{ eV}$

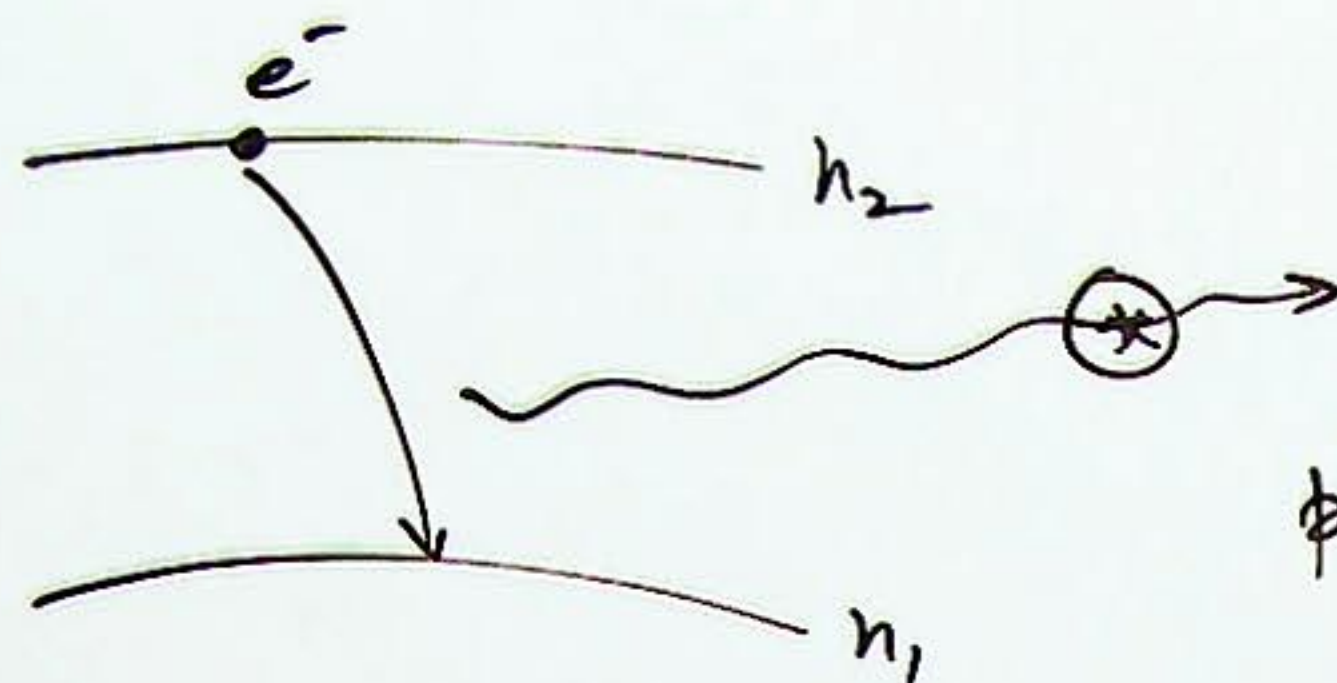
this photon will not be absorbed

⑥ Frequency & Wavelength of Emitted Radiation:

Vimp Alternative way to
Calculate λ :



$$\lambda = \frac{12431}{2.55} \text{ \AA}$$



$$\Delta E = \frac{hc}{\lambda}$$

(in eV)

$$\lambda = \frac{hc}{\Delta E(\text{in eV})} = \frac{12431}{\Delta E(\text{in eV})} \text{ \AA}$$

$$\Delta E(\text{in eV}) = \frac{12431}{\lambda(\text{in \AA})}$$

photon energy $\Delta E = E_{n_2} - E_{n_1} = \frac{hc}{\lambda}$

$$\left(-13.6 \frac{Z^2}{n_2^2}\right) - \left(-13.6 \frac{Z^2}{n_1^2}\right) = \frac{hc}{\lambda}$$

$$\Rightarrow \frac{1}{\lambda} = \frac{13.6}{hc} Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Rydberg's Formula.

$$\frac{1}{\lambda} = R Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

"wave no"

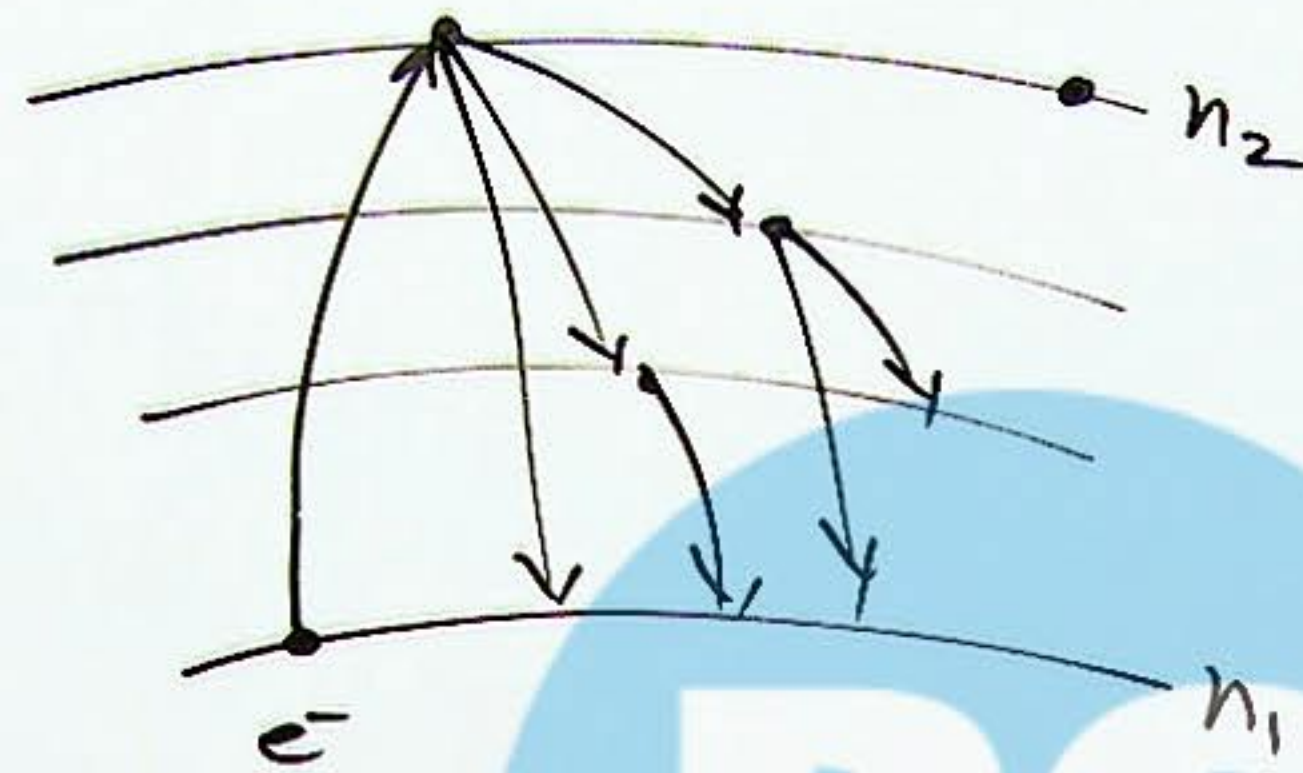
$$R = 10967800 \text{ m}^{-1}$$

$$\approx 10^7 \text{ m}^{-1}$$

⑦ De-excitation of an Atom:

After life time of excitation ($\sim 10^{-8}$ s)

an e^- transit to
any of lower
energy levels by
releasing photons



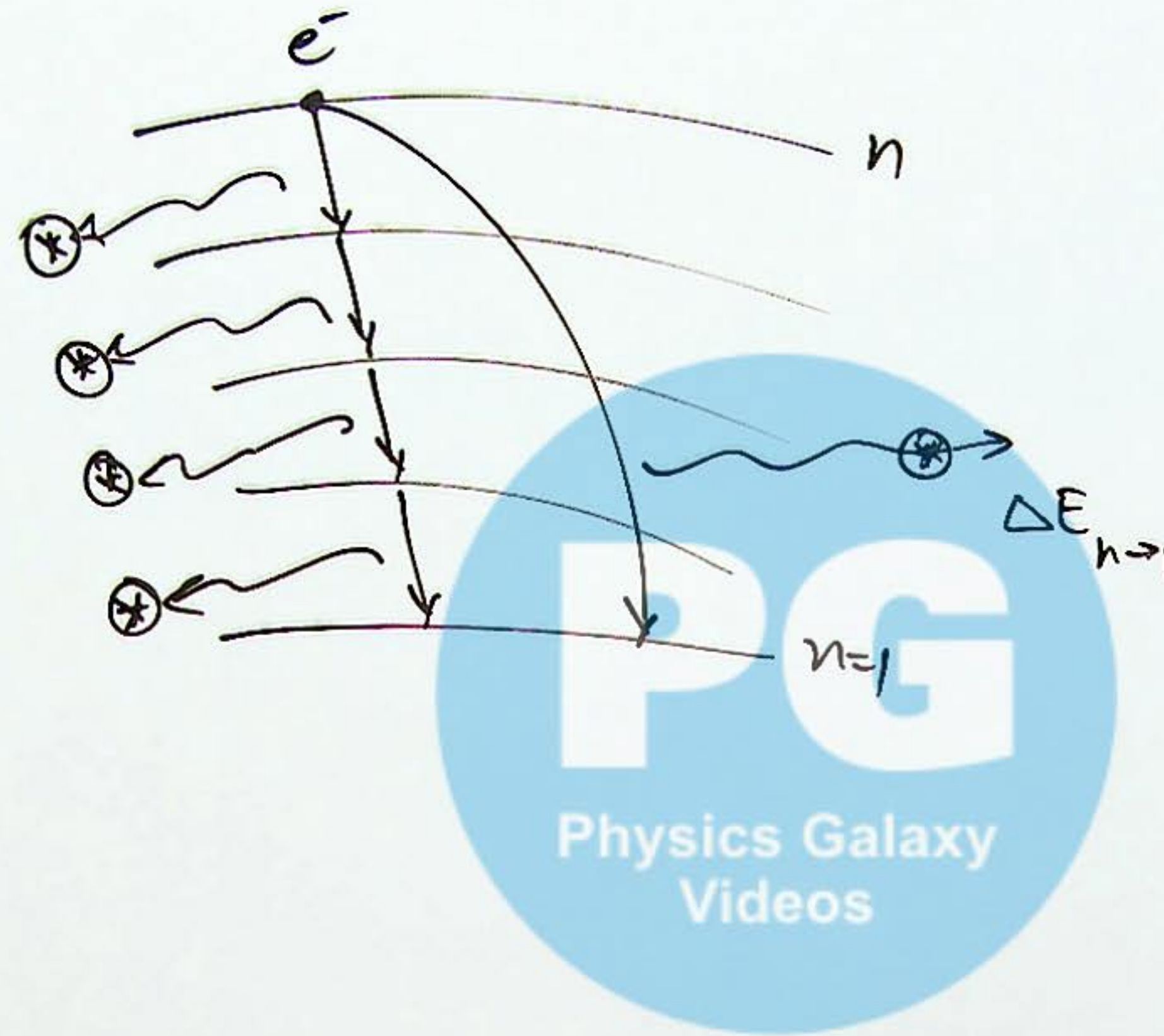
no of possible photon energies
emitted due to deexcitation from
 $n=n_2$ state are

$$\text{No of } \underline{\text{lines}} = {}^{n_2}C_2 = \frac{n(n-1)}{2}$$

Spectral.

⑧ Maximum Spectral lines emitted
by "One Atom" :

Max possible
photons emitted
by one Atom
 $= \underline{(n-1)}$



from state n , possible
spectral lines which can be
emitted $= {}^nC_2$

⑨ Hydrogen Spectrum & Spectral Series:

first line last line

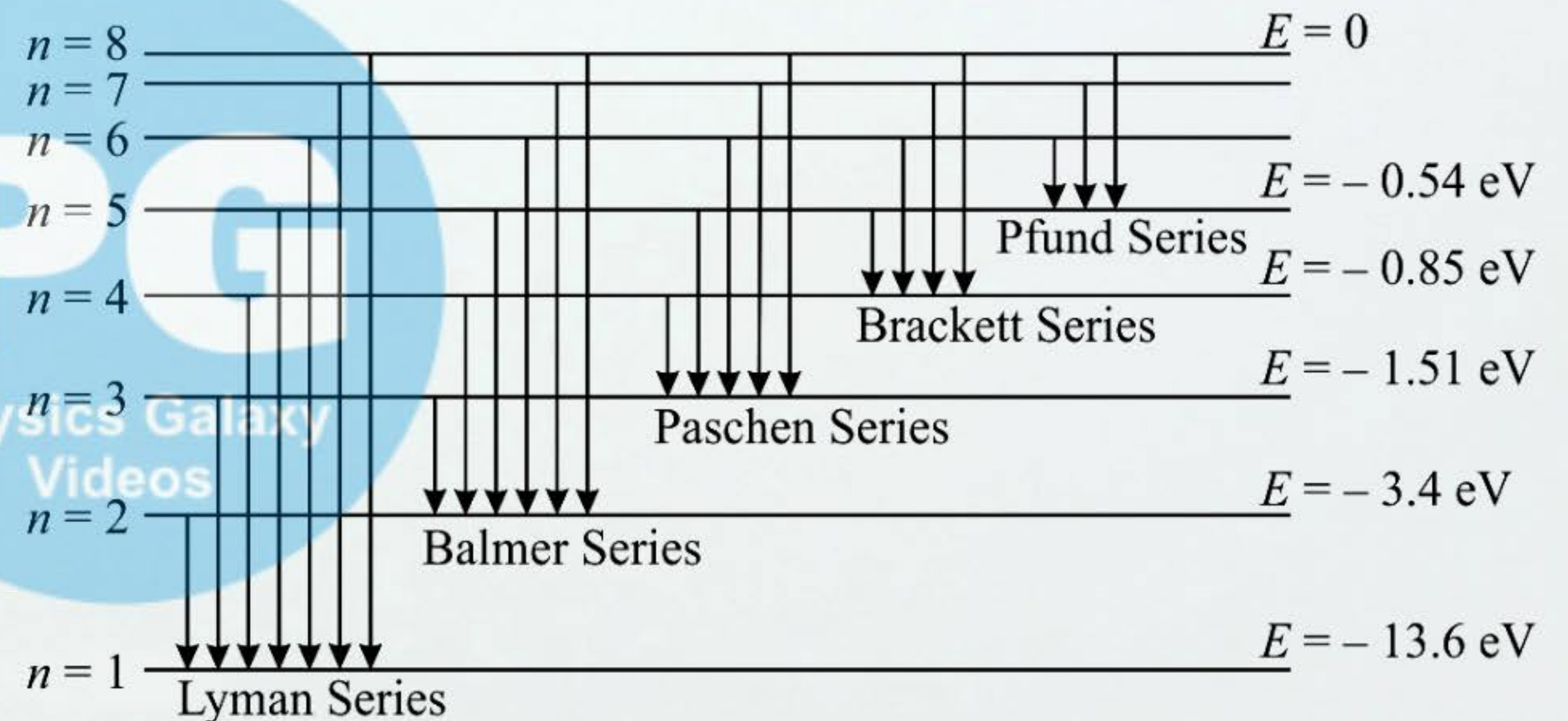
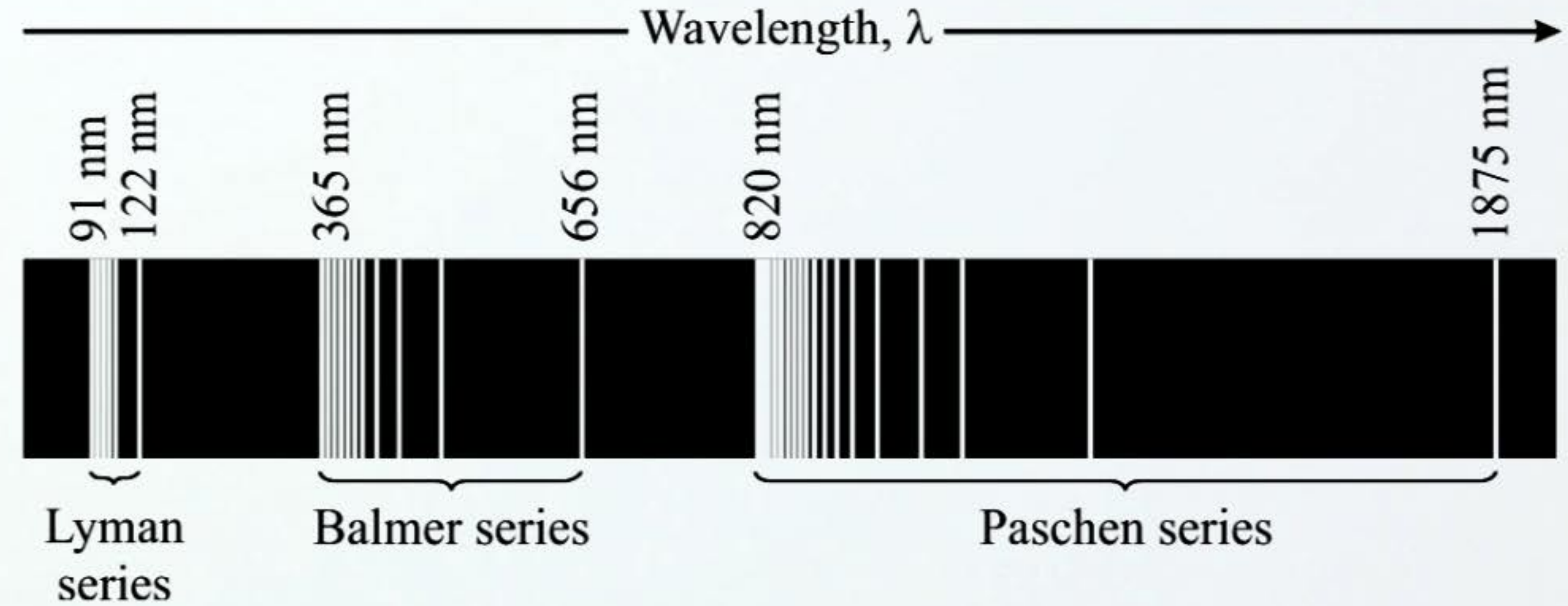
Lyman Series. FL LL
 $2 \rightarrow 1$ $\infty \rightarrow 1$

Balmer Series $3 \rightarrow 2$ $\infty \rightarrow 2$

' ' '

' ' '

' ' '



⑩ Effect of Nuclear mass on Bohr Model :

Diagram illustrating the effect of nuclear mass on the Bohr model of an atom.

Left Diagram (Bohr Model with Finite Nuclear Mass):

- A central nucleus is labeled $Nu(\text{rest})$ with a positive charge $+$.
- An electron e^- is shown in a circular orbit of radius r_n .
- The reduced mass μ_e is defined as $\mu_e = \frac{m_e m_N}{m_e + m_N}$.
- The Bohr radius for the n -th orbit is given by:

$$r_n = \frac{n^2 h^2 (m_e + m_N)}{4 \pi^2 K Z e^2 m_e m_N}$$
- The Bohr radius for an infinite nuclear mass is given by:

$$r_{n\infty} = \frac{n^2 h^2}{4 \pi^2 K Z e^2 m}$$
- The relationship between the two radii is shown in a box:

$$r_n = r_{n\infty} \times \left(\frac{m_e + m_N}{m_N} \right)$$

Right Diagram (Bohr Model with Infinite Nuclear Mass):

- The nucleus is represented by a central point labeled $\oplus \text{Com}$.
- Electrons e^- are shown in circular orbits.

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⑪ Use of Bohr model in different Potential Energy Systems:

IL: In a system a ^{revolving} charge has PE given as

$$U = k e \ln r$$

$$U(r) = f(r)$$

Using Bohr model find radius of n^{th} orbit for this charge.

Solⁿ: force on charge $f = -\frac{dU}{dr}$

$$|f| = \frac{ke}{r} = \frac{mv^2}{r}$$

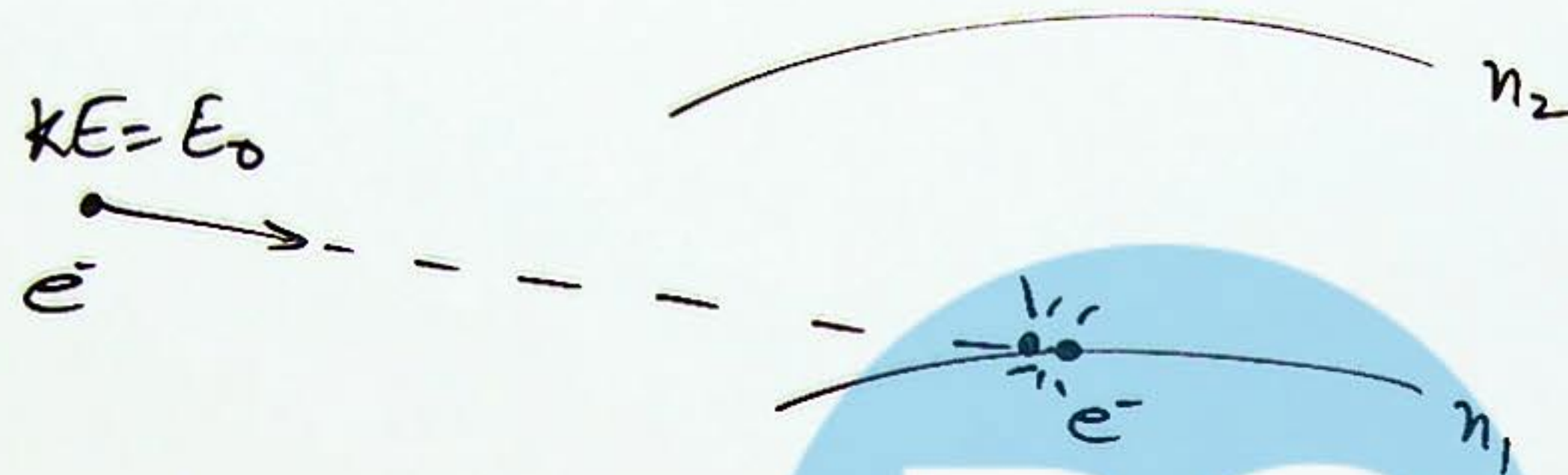
modified eqⁿ of P1.

$$\Rightarrow mv_n^2 = ke$$

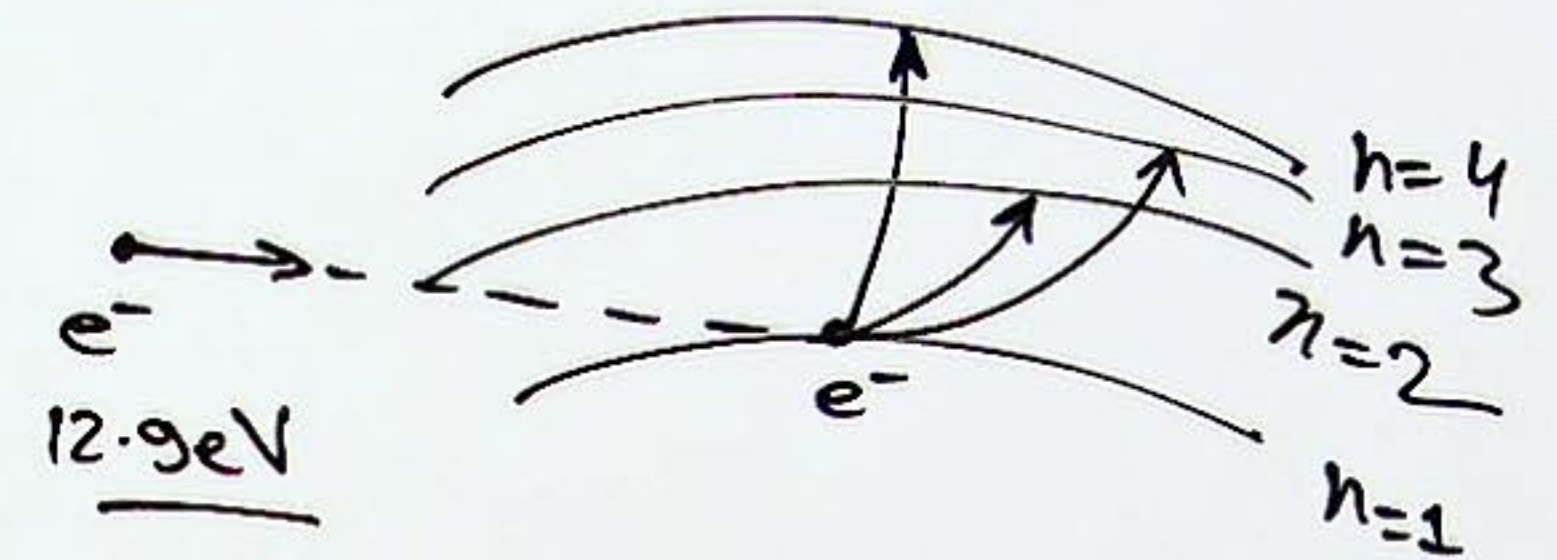
by P2 we have $mv_n r_n = \frac{nh}{2\pi}$

$$\Rightarrow r_n = \frac{nh}{2\pi m \sqrt{\frac{ke}{m}}} = \frac{nh}{2\pi \sqrt{mke}} \checkmark$$

⑫ Atomic Collisions:



NOTE: In above collision the supplied energy to atomic e^- can be any fraction of E_0



$$\Delta E_{14} = 12.75 \text{ eV} \quad \text{H-atom}$$